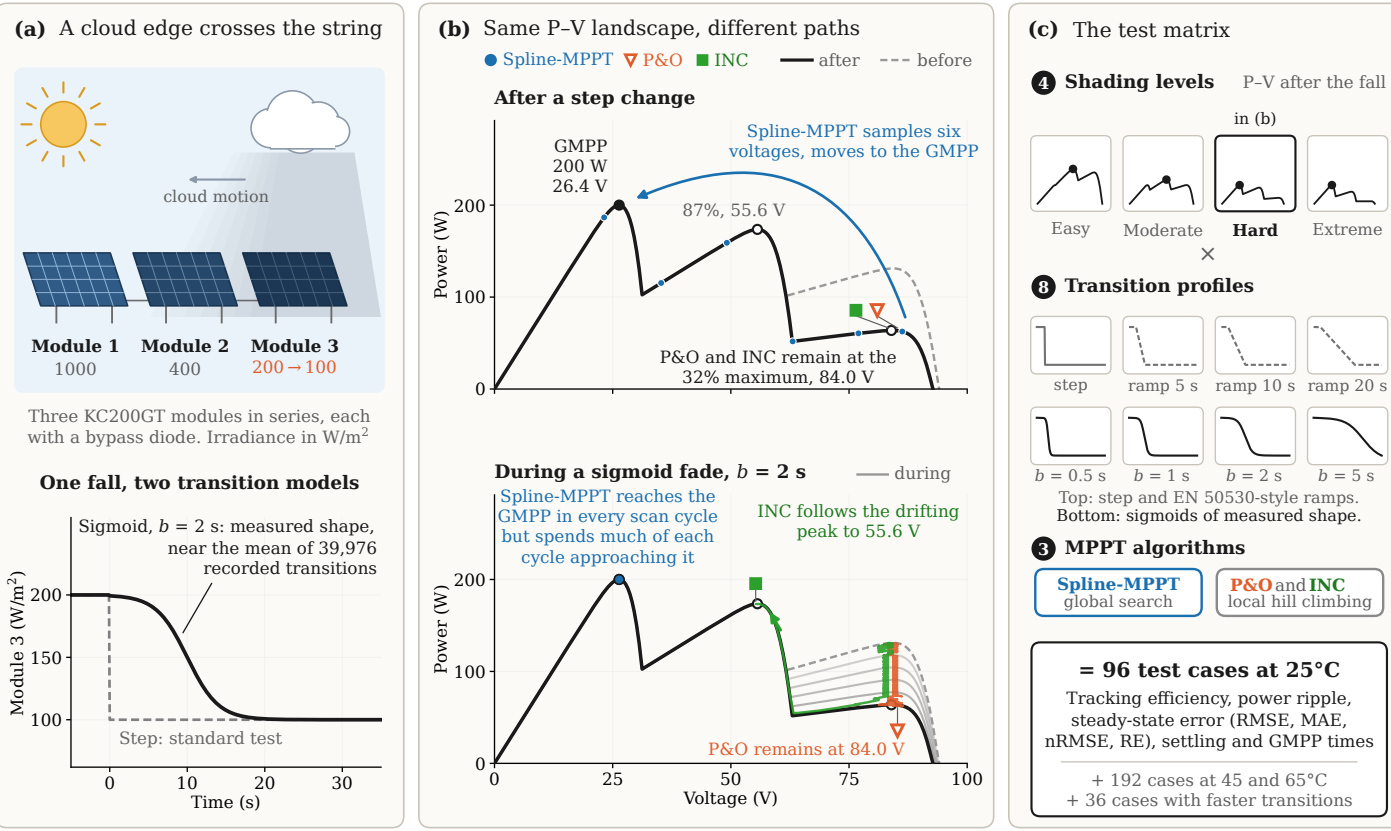

(a) A cloud edge crosses the string
cloud motion
Module 1
1000
Module 2
400
Module 3
200 → 100
Three KC200GT modules in series, each with a bypass diode. Irradiance in W/m²
One fall, two transition models
Sigmoid, b = 2 s: measured shape, near the mean of 39,976 recorded transitions
Step: standard test
Module 3 (W/m²)
Time (s)
(b) Same P–V landscape, different paths
Spline-MPPT
P&O
INC
after
before
After a step change
GMPP
200 W
26.4 V
Spline-MPPT samples six voltages, moves to the GMPP
87%, 55.6 V
P&O and INC remain at the 32% maximum, 84.0 V
Power (W)
During a sigmoid fade, b = 2 s
during
Spline-MPPT reaches the GMPP in every scan cycle but spends much of each cycle approaching it
INC follows the drifting peak to 55.6 V
P&O remains at 84.0 V
Voltage (V)
(c) The test matrix
4 Shading levels
P–V after the fall in (b)
Easy
Moderate
Hard
Extreme
×
8 Transition profiles
step
ramp 5 s
ramp 10 s
ramp 20 s
b = 0.5 s
b = 1 s
b = 2 s
b = 5 s
Top: step and EN 50530-style ramps.
Bottom: sigmoids of measured shape.
3 MPPT algorithms
Spline-MPPT global search
P&O and INC local hill climbing
= 96 test cases at 25°C
Tracking efficiency, power ripple, steady-state error (RMSE, MAE, nRMSE, RE), settling and GMPP times
+ 192 cases at 45 and 65°C
+ 36 cases with faster transitions